Q1. Student Performance Analysis

You have a table Student(sid, name, marks, dept).

Write SQL queries to:

1. Find the **second highest marks**.
2. List students scoring **above department average**.
3. Display department-wise **top scorer**.

Q2. Employee Salary Management

Table: Employee(emp_id, name, salary, dept_id)

Write queries to:

1. Find employees earning **more than average salary**.
2. Show **department-wise total salary**.
3. List employees whose salary is **greater than all employees in dept 10**.

Q3. E-commerce Orders

Tables:

Customer(cid, cname)

Orders(oid, cid, amount)

Write queries to:

1. Find customers who **never placed any order**.
2. Find customers with **maximum order amount**.
3. Show customers with **total purchase > 5000**.

Q4. Join-Based Scenario

Tables:

Student(sid, name, dept_id)

Department(dept_id, dept_name)

Write queries to:

1. Display **student name with department name**.
2. Find students **not assigned to any department**.
3. Count students **in each department**.

Q5. Aggregate + HAVING

Table: Sales(sale_id, product, quantity, price)

Write queries to:

1. Find total sales **per product**.
2. Show products with **total sales > 10000**.
3. Display product with **maximum revenue**.

Q6. Subquery Scenario

Table: Employee(emp_id, name, salary)

Write queries to:

1. Find employees with **second highest salary**.
2. Find employees earning **more than minimum salary**.
3. List employees whose salary is **equal to any manager salary** (assume manager table).

Q7. Set Operations

Tables:

TableA(id)

TableB(id)

Write queries to:

1. Find **common records** (INTERSECT logic).
2. Find records **present in A but not in B**.
3. Combine both tables **without duplicates**.

Q8. Constraints & Table Design

Write SQL to:

1. Create table Orders with
    - order_id as PRIMARY KEY
    - cust_id as FOREIGN KEY
    - amount NOT NULL
2. Add UNIQUE constraint on cust_id.
3. Modify table to add column order_date.

Q9. View & Index Scenario

Table: Employee(emp_id, name, dept, salary)

Write queries to:

1. Create a **view** showing employees with salary > 50000.
2. Create an **index on salary column**.
3. Display data from the view.

Q10. Transaction Control

Write SQL statements to:

1. Start a transaction and insert data.
2. Update a record and **ROLLBACK** changes.
3. Insert again and **COMMIT**.